

Foundation Model II

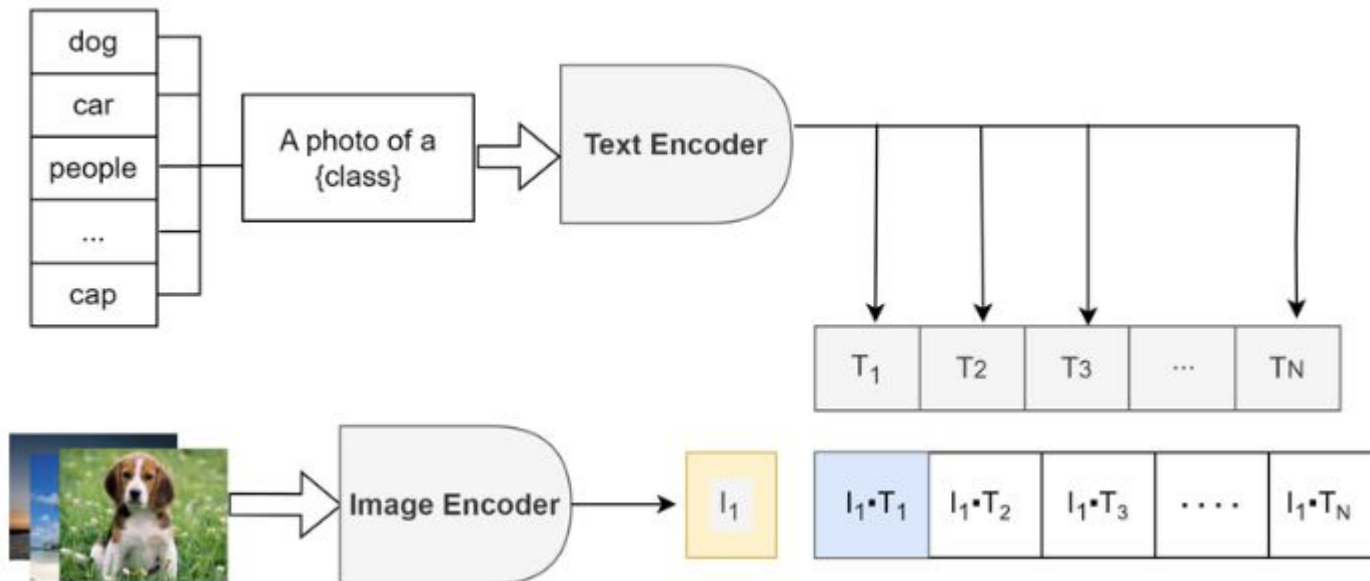
Guosheng Hu

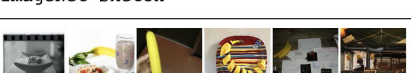
Foundation Model

Domain	Foundation Model	Description
Language	GPT (OpenAI), LLaMA (Meta), Claude (Anthropic)	Trained on large text corpora
Vision	DINO, SAM (Meta)	Trained on images
Vision–Language	CLIP , LLaVA, Stable Diffusion	Combine visual and textual understanding
Biology	AlphaFold, ESM	Trained on protein sequences and structures

CLIP - Introduction

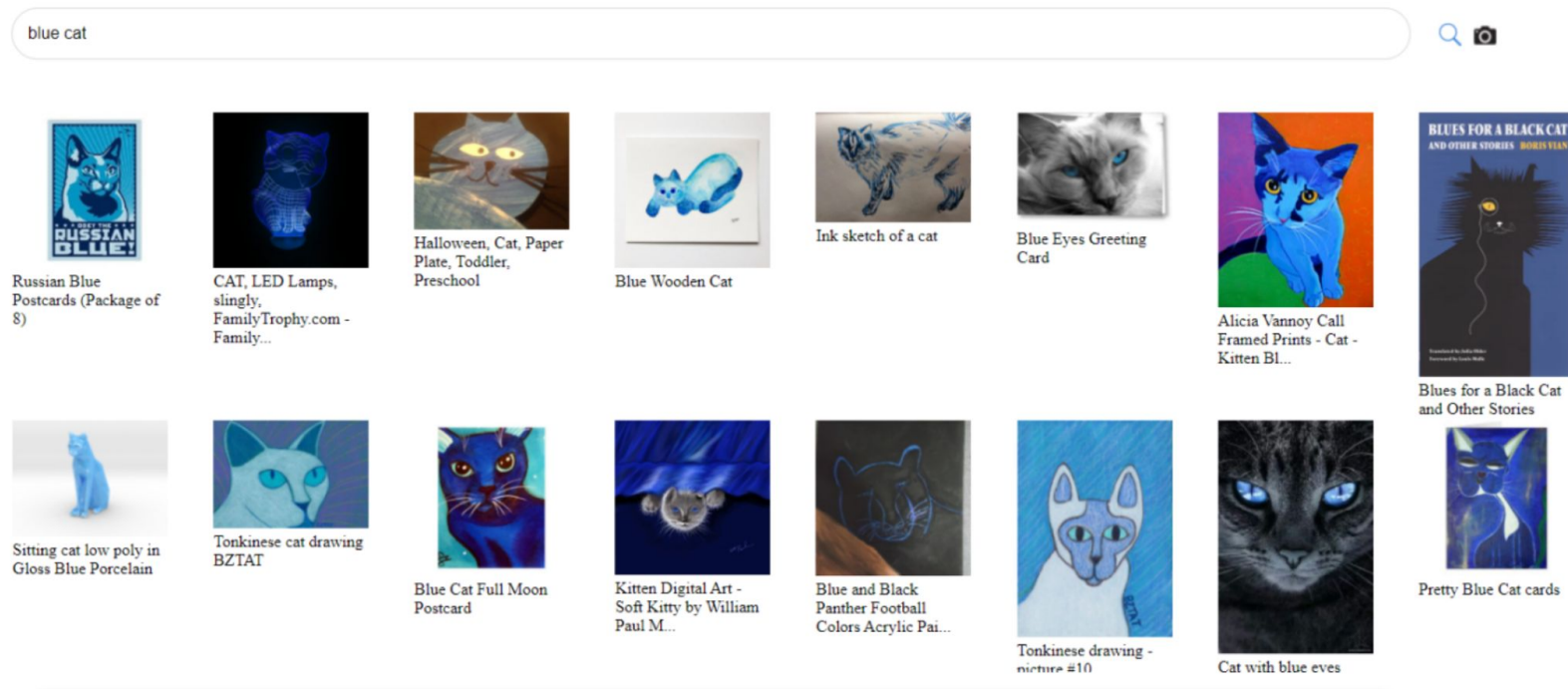
Contrastive Language-Image Pre-Training (CLIP) is a model that learns the **connection between images and natural language text**. It embeds both images and text into the **same high-dimensional space**. This allows it to perform **zero-shot tasks**, like classifying an image based on a text description, **without** direct training for that specific task.



Dataset	ImageNet	
	ResNet101	CLIP ViT-L
 ImageNet	76.2%	76.2%
 ImageNet V2	64.3%	70.1%
 ImageNet Rendition	37.7%	88.9%
 ObjectNet	32.6%	72.3%
 ImageNet Sketch	25.2%	60.2%
 ImageNet Adversarial	2.7%	77.1%

CLIP - Training Data

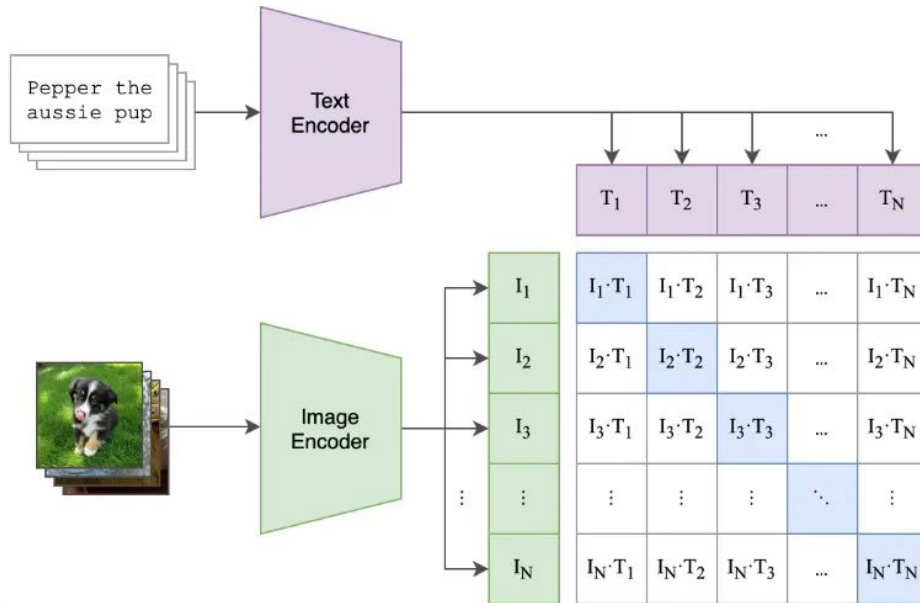
CLIP was trained on a massive, custom-built dataset of **400 million (image, text) pairs** collected directly from the internet. The text captions were **not perfectly curated**, representing a wide, noisy, and natural distribution of language. This **large-scale, "weakly supervised"** dataset is the key to its **robust and generalizable** understanding of visual concepts.



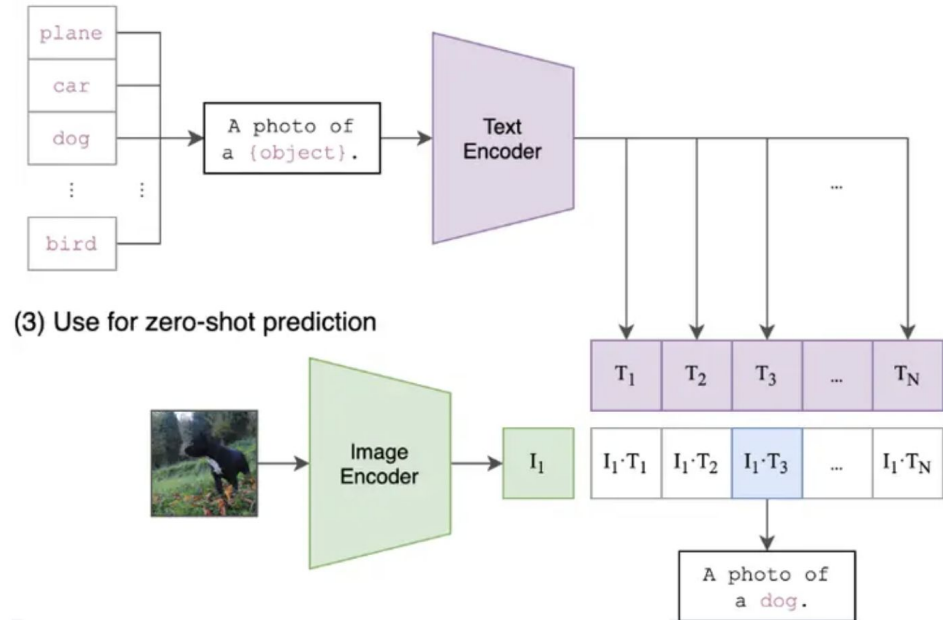
CLIP - Training (Arch. and Loss)

The architecture consists of two separate encoders: an **image encoder** (like a Vision Transformer) and a **text encoder** (a text Transformer). During training, the model **maximises** the **similarity** of the embeddings for **correct (image, text) pairs** while **minimising** the similarity for **incorrect pairs**. This is achieved using a "**contrastive loss**" function, which effectively learns to **match images with their corresponding descriptions** from a large batch.

(1) Contrastive pre-training

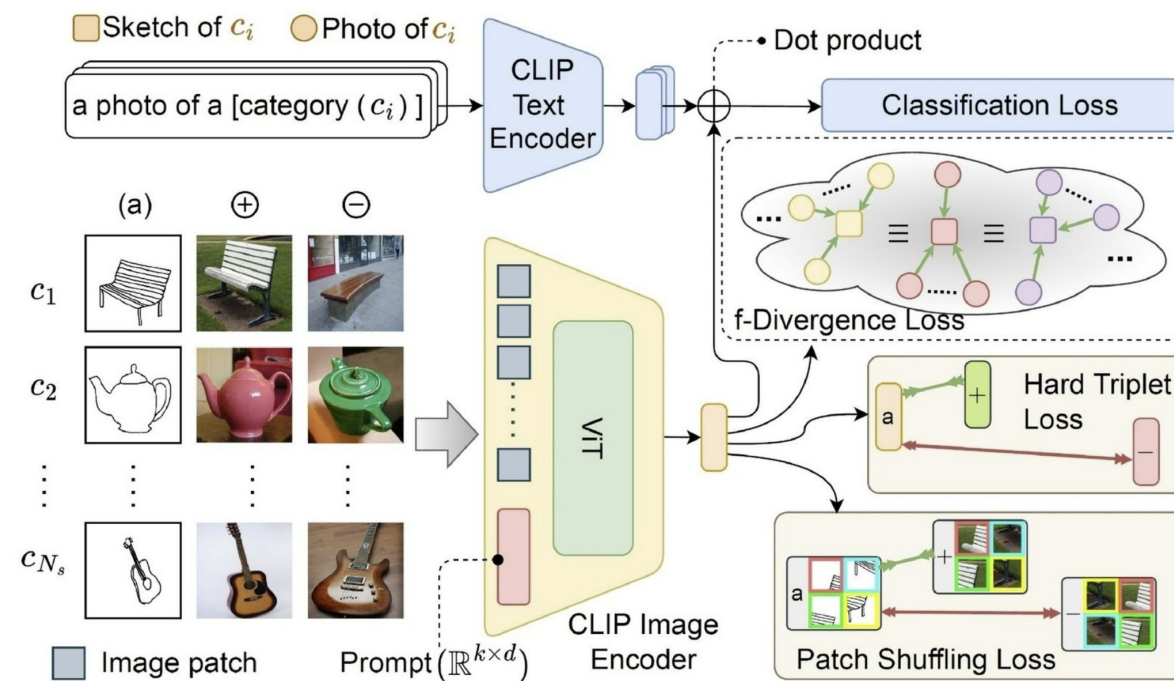
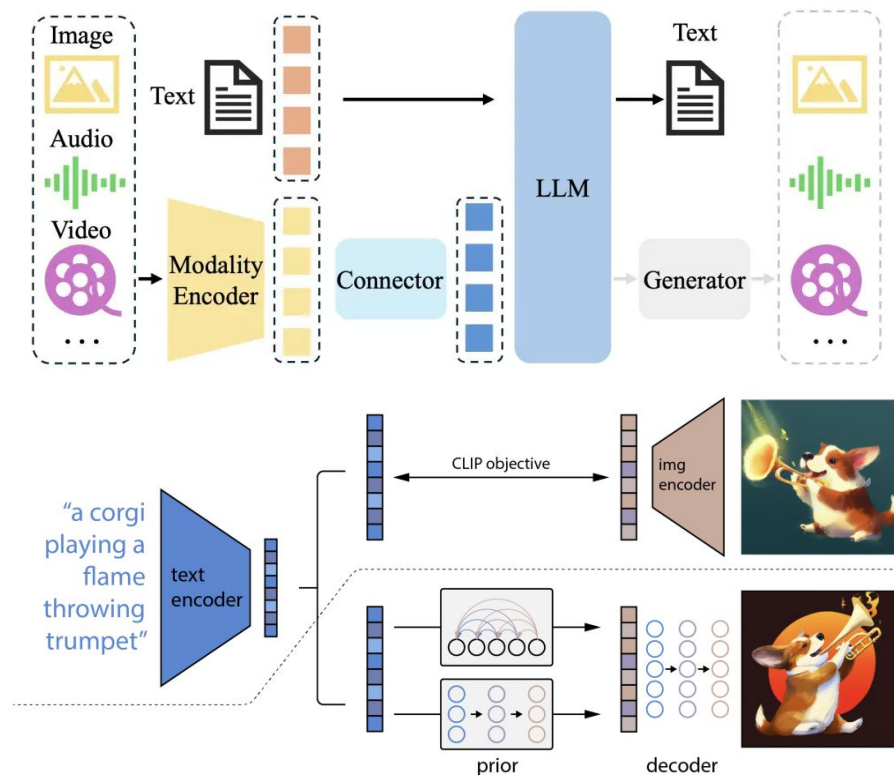


(2) Create dataset classifier from label text



CLIP - Applications

CLIP is famously used as the **core component** for **generative models** like DALL-E 2, guiding the image synthesis to match a text prompt. It also enables highly effective **natural language image retrieval** and **multimodal large language models**.



Two Modalities to More: ImageBind



Web Image-Text



Sheep basking in the sun

Depth Sensor Data



Web Videos



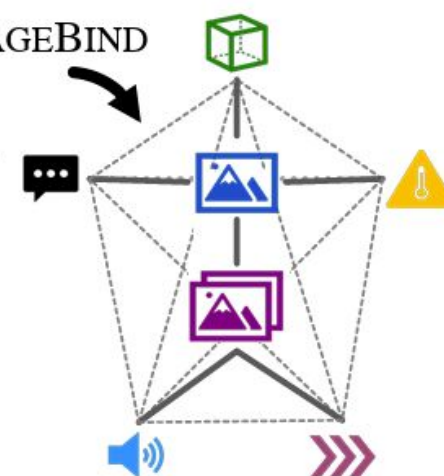
Thermal Data



Egocentric Videos



IMAGEBIND



Girdhar, Rohit, et al. "Imagebind: One embedding space to bind them all." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2023.

Applications

1) Cross-Modal Retrieval

Audio



Crackle of a Fire

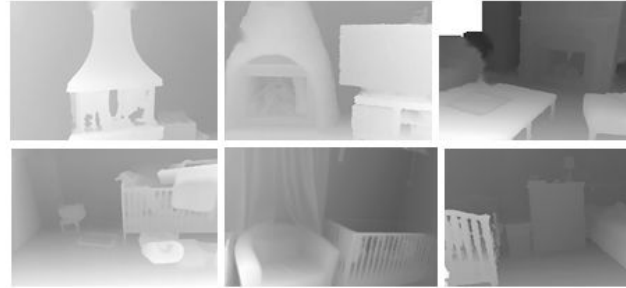


Baby Cooing

Images & Videos



Depth



Text

"A fire crackles while a pan of food is frying on the fire."

"Fire is crackling then wind starts blowing."

"Firewood crackles then music..."

"A baby is crying while a toddler is laughing."

"A baby is laughing while an adult is laughing."

"A baby laughs and something..."

2) Embedding-Space Arithmetic



Waves



3) Audio to Image Generation



Dog



Engine



Fire



Rain



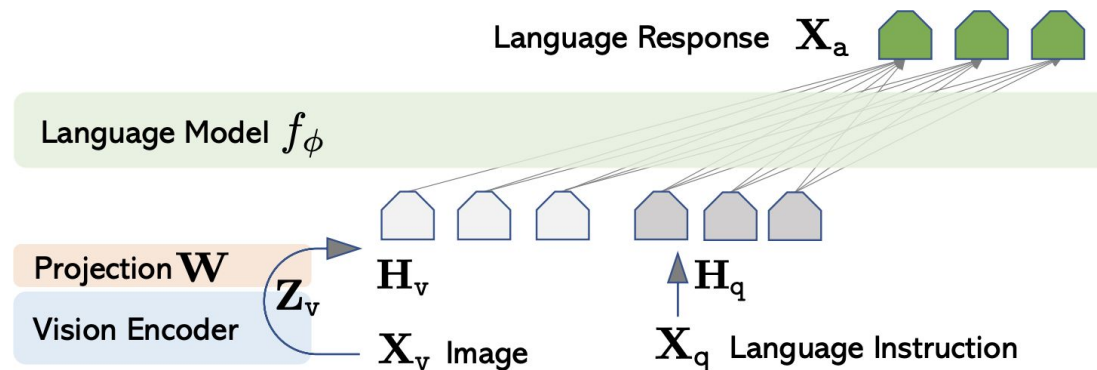
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LLaVA - Introduction

Large Language and Vision Assistant (LLaVA), is a **multimodal large language model** designed for **general-purpose** visual and language understanding. It connects a **pre-trained vision encoder** (like CLIP's) with a **large language model** (like Vicuna). This architecture enables LLaVA to understand both images and text, allowing for sophisticated visual **reasoning** and **conversation**.



Visual input example, Extreme Ironing:



Source: <https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg>

User
LLaVA

What is unusual about this image?

The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment.

User
LLaVA

[Start a new conversation, and clear the history]

What's happening in the scene?

The image depicts a man standing on top of a yellow SUV in a busy city street. He is holding a portable ladder, seemingly ironing clothes while standing on the vehicle. Around the scene, there are other cars, a traffic light, and a fire hydrant. Additionally, there are two other people in the vicinity, one close to the man on the car and another further away. The man on the ladder appears to be attracting attention **due to his unconventional choice of ironing his clothes on top of a moving car**. The city street around him is bustling with activity, adding to the unique nature of the scene.

LLaVA - Training Data

LLaVA was trained using a novel and data-efficient "**visual instruction tuning**" dataset. This dataset was created by using **GPT-4** to **generate detailed conversations, questions, and reasoning steps** based on images.

- "Describe the following image in detail"
- "Provide a detailed description of the given image"
- "Give an elaborate explanation of the image you see"
- "Share a comprehensive rundown of the presented image"
- "Offer a thorough analysis of the image"
- "Explain the various aspects of the image before you"
- "Clarify the contents of the displayed image with great detail"
- "Characterize the image using a well-detailed description"
- "Break down the elements of the image in a detailed manner"
- "Walk through the important details of the image"
- "Portray the image with a rich, descriptive narrative"
- "Narrate the contents of the image with precision"
- "Analyze the image in a comprehensive and detailed manner"
- "Illustrate the image through a descriptive explanation"
- "Examine the image closely and share its details"
- "Write an exhaustive depiction of the given image"

Table 12: The list of instructions for detailed image description.

Context type 1: Captions

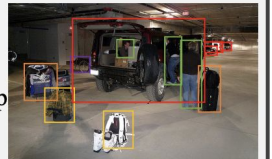
A group of people standing outside of a black vehicle with various luggage.

Luggage surrounds a vehicle in an underground parking area

People try to fit all of their luggage in an SUV.

The sport utility vehicle is parked in the public garage, being packed for a trip

Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation

Question: What type of vehicle is featured in the image?

Answer: The image features a black sport utility vehicle (SUV) ...<omitted>

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>

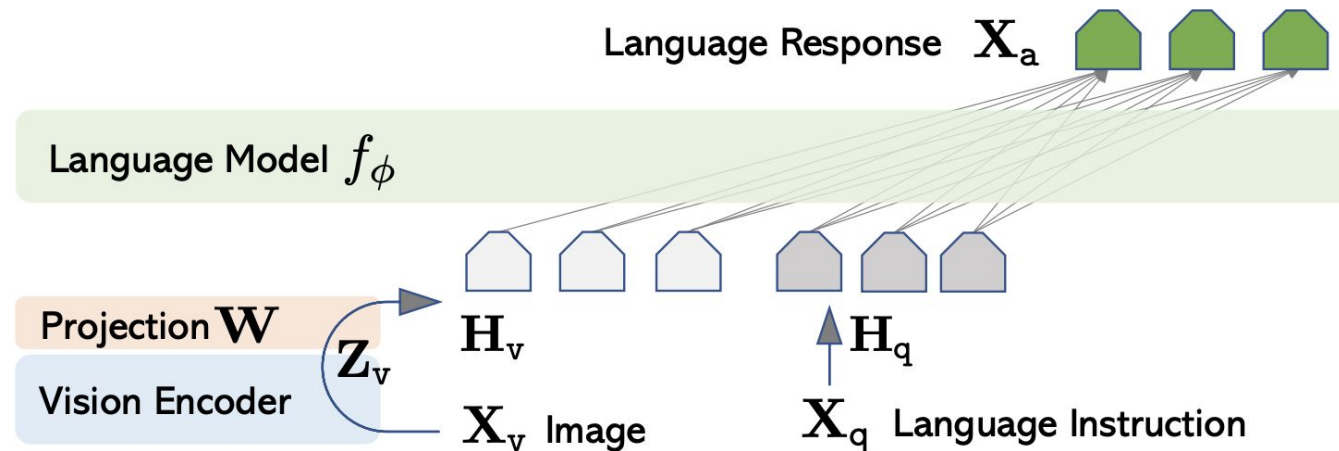
Response type 3: complex reasoning

Question: What challenges do these people face?

Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

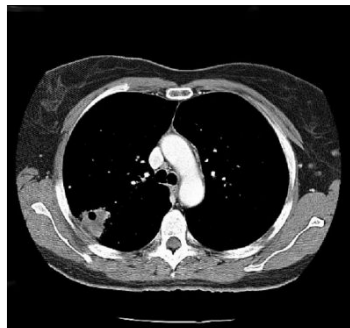
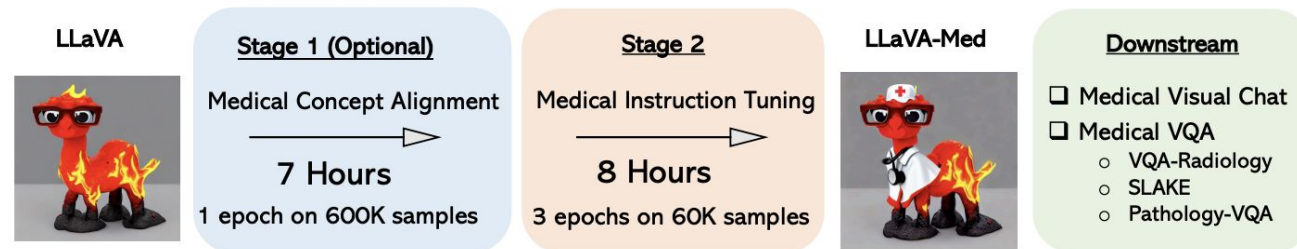
LLAVA - Training (Arch. and Loss)

The architecture features a **frozen vision encoder** and a **frozen LLM**, connected by a **simple, trainable projection matrix $\mathbf{W}$** (an MLP). Training is a two-stage process: first, the projection matrix is trained to **align image features with the LLM's word embedding space**. Second, the model undergoes **end-to-end fine-tuning** on the visual instruction data, adapting both the LLM and the projection matrix.



LLaVA - Applications

LLaVA excels at conversational AI, allowing users to have a **detailed dialogue about an image**, ask **complex questions**, and **request descriptions**. This makes it a powerful foundation for enhanced **visual question answering (VQA) systems** and **accessible, image-aware chatbots** in different areas like medical or remote-sensing.



Conversation:

User: Where is the lesion located?

Assistant: The lesion is located in the posterior aspect of the right upper lobe of the lung.

User: Is it extending into the chest wall?

Assistant: Yes, the CT scan suggests that the large cavitating lesion in the right upper lobe is likely extending into the chest wall.

User: Can you explain more about the cavitating lesion?

Assistant: Certainly. A cavitating lesion refers to an abnormal mass or growth within the lung tissue that has a cavity or hollow space inside it. This cavity can be formed due to the necrosis (death) of the tissue inside the lesion, leading to the breakdown and disintegration of the affected tissue. Cavitating lesions can be caused by various factors, such as infections (like tuberculosis or fungal infections), autoimmune diseases, or malignancies (cancer). In the context of the CT scan provided, the large cavitating lesion in the right upper lobe of the lung is suggestive of a malignant (cancerous) growth. The cavity within the lesion may be a result of the rapid and uncontrolled growth of cancer cells, which can lead to the destruction of surrounding healthy tissue. It's important to note that further diagnostic tests, such as a biopsy, would be required to confirm the malignancy and determine the specific type of cancer involved.

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Diffusion Model

Motivation

Can we generate data by **gradually constructing it**, instead of producing it all at once?

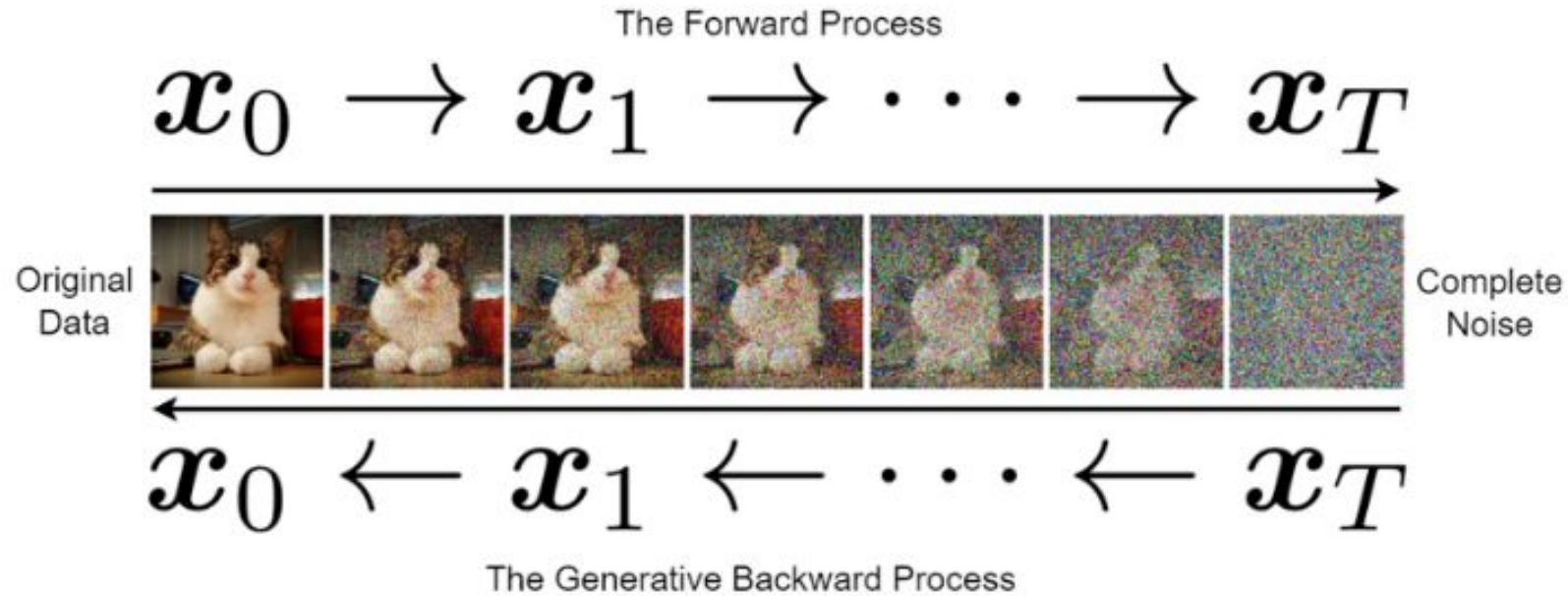
Applications

- Image Generation (DALL-E, Stable Diffusion, Gemini, etc)
- Video Generation (Sora, Gemini, Seedance 2.0, etc)

Classical approaches

- GANs → unstable training
- VAEs → blurry images
- Autoregressive models → slow generation

Forward (Destroy) and Backward (Construct)



Benyamin Ghojogh, Ali Ghodsi, Diffusion Models: Tutorial and Survey, 2024

Learning to Reverse Noise

Forward Process (No Learning)

Take a real image $x_0 \sim p_{\text{data}}(x)$

Add small Gaussian noise repeatedly $x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1 - \alpha_t} \epsilon$

where $\epsilon \sim \mathcal{N}(0, I)$ (Gaussian noise), $\alpha_t \in (0, 1)$

After many steps, we get pure noise $x_T \sim \mathcal{N}(0, I)$

Reverse Process (Learning)

Now we train a neural network $\epsilon_\theta(x_t, t)$ to predict the noise added at each step, not predict the noisy or clean image. Given the true noise ϵ , the loss is $\mathcal{L} = \mathbb{E} [\|\epsilon - \epsilon_\theta(x_t, t)\|^2]$

Sampling (Generating an Image)

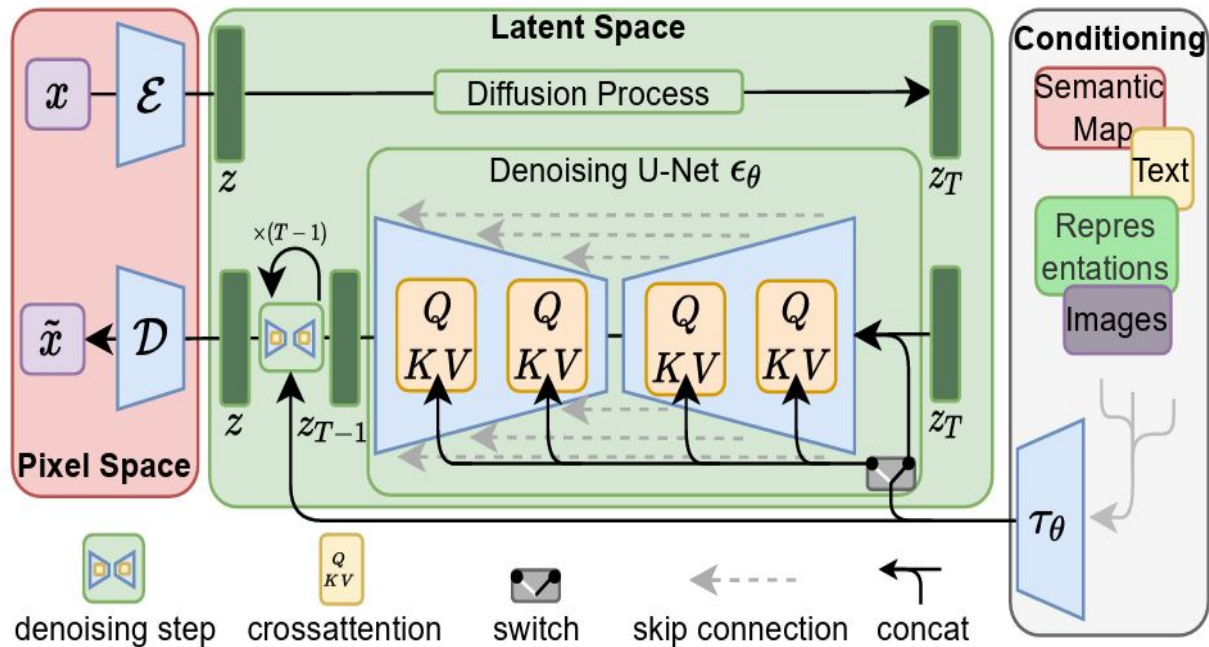
After training, we start from a noise $x_T \sim \mathcal{N}(0, I)$

We predict the noise $\hat{\epsilon} = \epsilon_\theta(x_t, t)$

Denoise $x_{t-1} = \text{denoise}(x_t, \hat{\epsilon})$

After many steps, we get x_0 (generated image)

Conditional Image Generation



Diffusion Process in **latent space**.
When **condition** (text, image) y is added,
then the neural network becomes:

$$\epsilon_\theta(z_t, t, y)$$

To achieve this, the condition is encoded
by $\tau_\theta(y)$, then the cross attention is used
to fuse image feature (Q) and condition
feature (KV).

Rombach, Robin, et al. "High-resolution image synthesis with latent diffusion models." *CVPR*. 2022.

Conditional Image Generation

Input (Canny Edge)



Default



Automatic Prompt



“a man with beard sitting with two children”

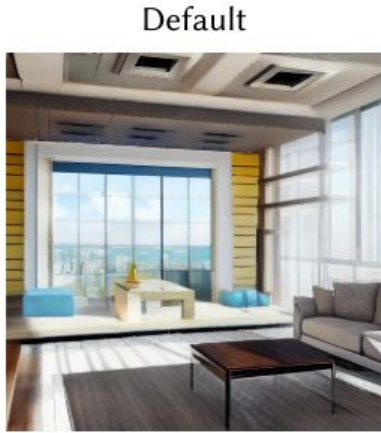
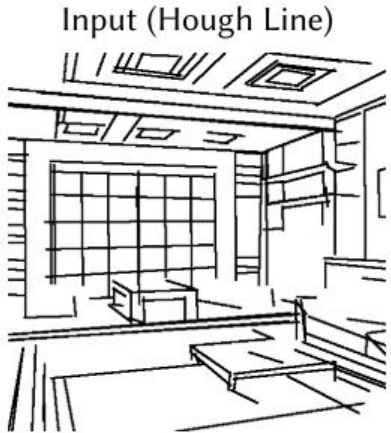
User Prompt



“mother and two boys in a room, masterpiece, artwork”

Zhang, Lvmin, Anyi Rao, and Maneesh Agrawala. "Adding conditional control to text-to-image diffusion models." *ICCV*. 2023.

Conditional Image Generation



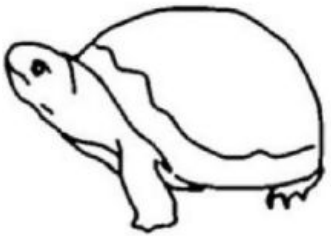
“a living room with a couch and a window”



“a fantastic living room made of wood”

Conditional Image Generation

Input (User Scribble)



Default



Automatic Prompt



“a turtle in river”

User Prompt



“a masterpiece of cartoon-style turtle illustration”

Conditional Image Generation

Input (HED Edge)



Default



Automatic Prompt



“a painting of a woman”

User Prompt



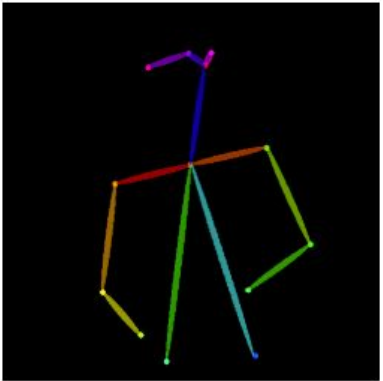
“... in cyan dress”



“... in red dress”

Conditional Image Generation

Input (openpose)



Default



User Prompt

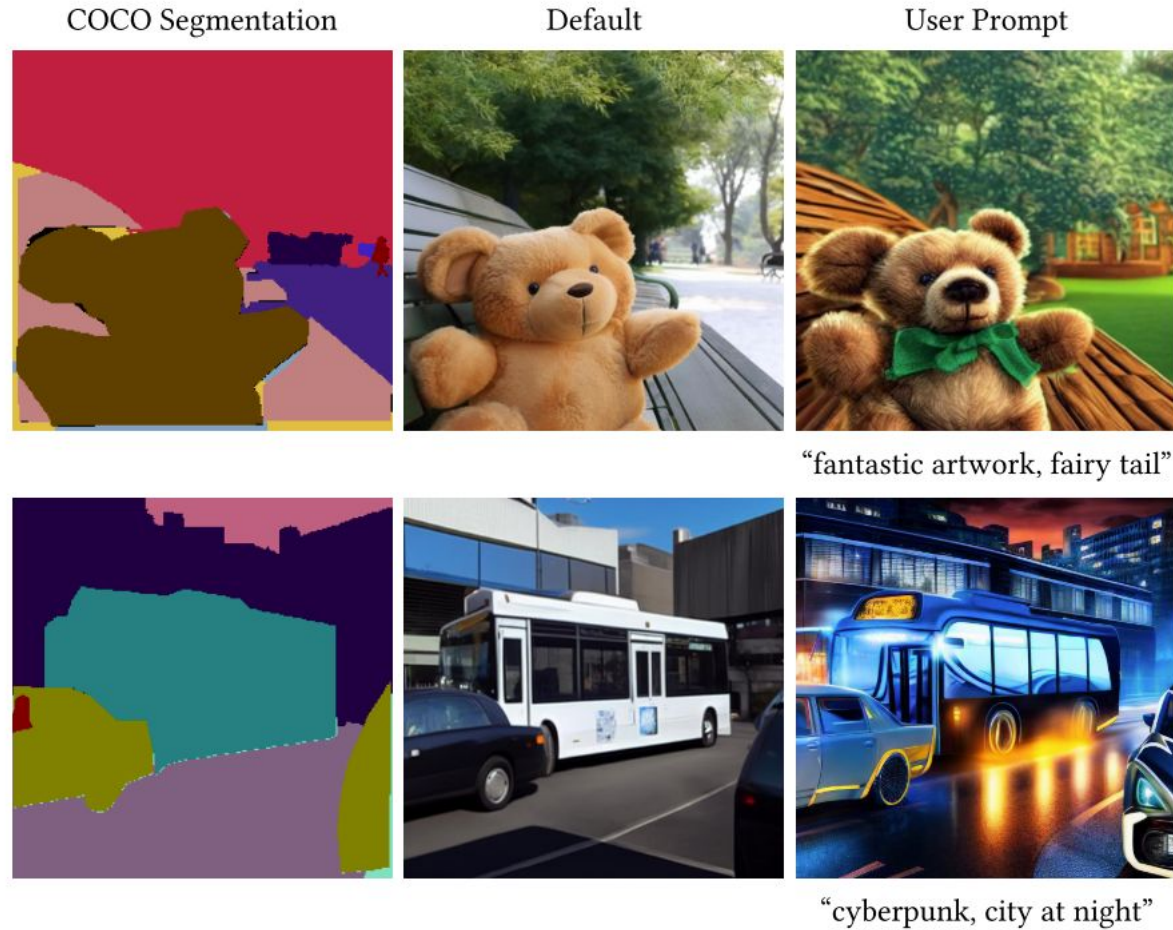


“chef in the kitchen”

Conditional Image Generation



Conditional Image Generation



Conditional Image Generation



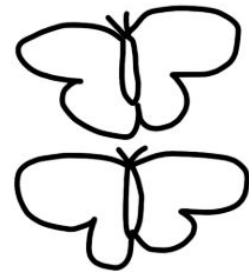
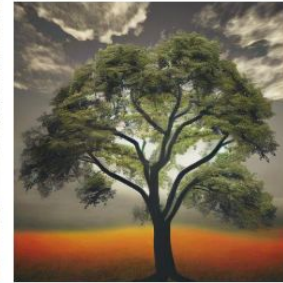
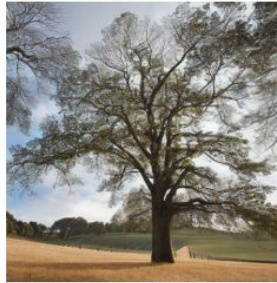
User input



Conditional Image Generation



without user prompt



without user prompt



“house”



What is the Next?

From Digital to Physical

Foundation Models in Physical World

Feature	Robotics (Traditional)	Embodied AI
Primary Focus	Hardware, control theory, kinematics, and precision.	Perception, reasoning, and learning from physical interaction.
Environment	Structured (factories, assembly lines).	Unstructured (homes, hospitals, outdoors).
Programming	Hard-coded instructions (e.g., Move joint A by 30°).	Learned behaviors (e.g., "Go find the coffee mug").
Goal	Repeatability and accuracy.	Generalization and adaptability to new tasks.

Foundation Models in Physical World

Feature	World Model	Embodied AI
Role	The Simulator (Understanding Physics).	The Agent (Making Decisions).
Input/Output	State + Action -> Next State (s, a -> s').	Vision/Goal -> Action (v, g -> a).
Analog	"How the world works."	"How I should behave."

Foundation Models in Physical World

Category	Foundation Model	Key Capability
World Models	JEPA (Meta), Dreamer, GAIA-1 (Wayve)	Predicts physical outcomes (s, a -> s')
Embodied AI	RT-2, OpenVLA, Figure-01	VLA Directly maps visual observations and text commands to robotic actions.